GENAI INTEGRATED PROJECT

AI Text Summarizer

Project Documentation Report

Project type: Generative AI Web Application

Core technology: Python + Streamlit + Google Gemini API

Purpose: Convert long text into concise summaries in different formats.

1. Abstract

The GenAI Text Summarizer is a simple web application that uses Generative Artificial Intelligence to convert long text into shorter, readable summaries. It demonstrates how an AI model can be integrated into a software application.

2. Problem Statement

Reading lengthy articles, study notes, and informational documents can take considerable time. Users may need a shorter version that preserves the main ideas.

3. Objectives

• Build a simple interface for entering text.
• Connect the application to a Generative AI model through an API.
• Allow users to choose the summary format and length.
• Display and download the generated summary.

4. Tools and Technologies

Python is used for application logic. Streamlit creates the web interface. The Google Gemini API provides generative AI capabilities, and the Google Gen AI Python SDK connects the application to the model.

5. System Workflow

1. The user pastes text into the application.
2. The user chooses a summary style and length.
3. The application sends the text and instructions to Gemini.
4. Gemini generates a summary.
5. The app displays the result and provides a download option.

6. GenAI Integration

The project reads the API key from the GEMINI_API_KEY environment variable. It sends the source text and formatting instructions to the gemini-2.5-flash model using the Google Gen AI Python SDK. The prompt asks the model to preserve important facts and avoid inventing details.

7. Sample Input

Renewable energy comes from sources that are naturally replenished, such as sunlight, wind, water, and geothermal heat. Solar panels convert sunlight into electricity, while wind turbines use moving air to generate power. Renewable energy can reduce greenhouse gas emissions and dependence on fossil fuels. However, some sources depend on weather conditions, and projects may require suitable locations, storage systems, and investment.

8. Illustrative Sample Output

• Renewable energy comes from naturally replenished sources such as sunlight, wind, water, and geothermal heat.
• Solar panels and wind turbines convert natural energy into electricity.
• Benefits include lower greenhouse gas emissions and reduced dependence on fossil fuels.

This is an illustrative sample, not a live API response.

9. Advantages

• Saves time when reviewing long text.
• Offers multiple summary formats and lengths.
• Provides a simple user interface.
• Demonstrates practical API integration.

10. Limitations

The application requires an internet connection and a valid API key. AI summaries may omit context or make mistakes. Users should verify important information against the original source.

11. Future Enhancements

PDF and DOCX upload, multilingual summaries, summary history, user accounts, and citation-aware summaries can be added in future versions.

12. Conclusion

The GenAI Text Summarizer demonstrates how Generative AI can be integrated into a Python web application. It combines a user-friendly interface with an AI API to generate summaries in different formats and lengths.